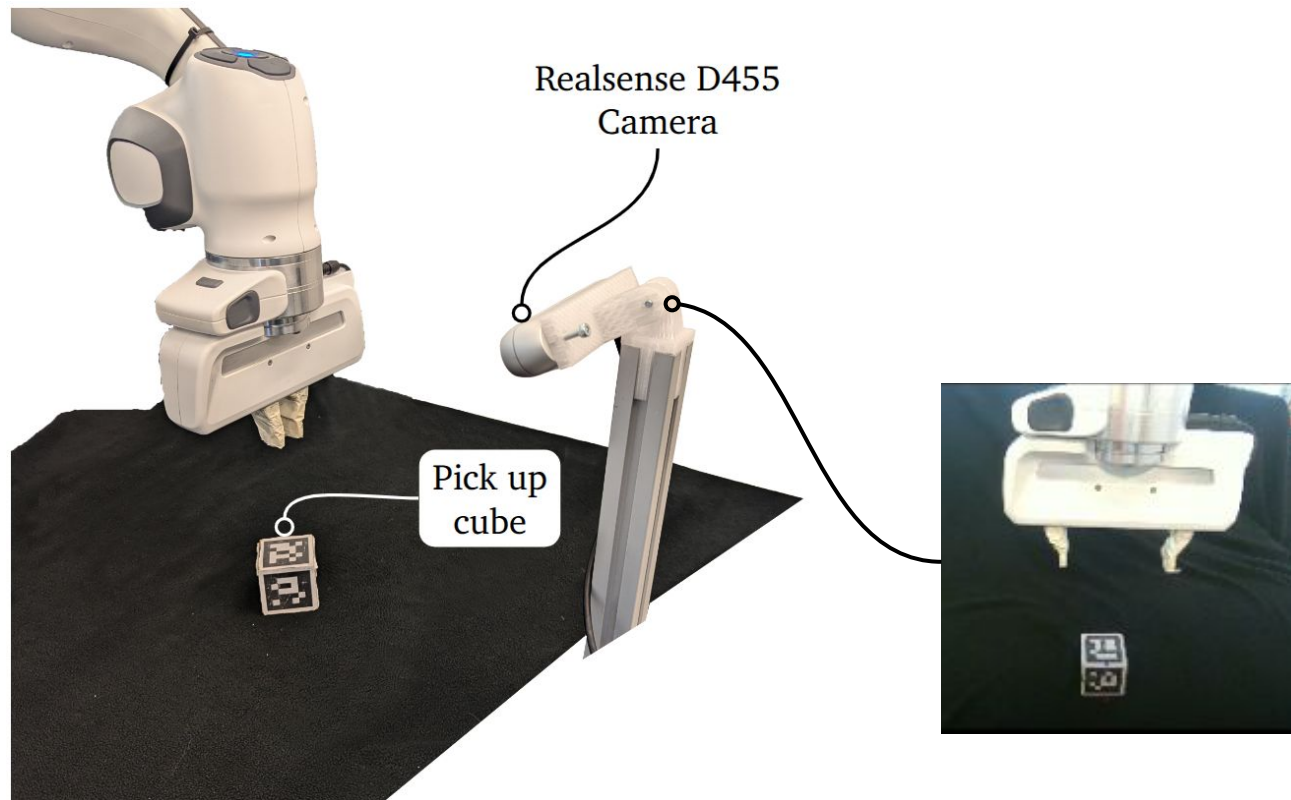


Yarden <> Nicolas

Introduction

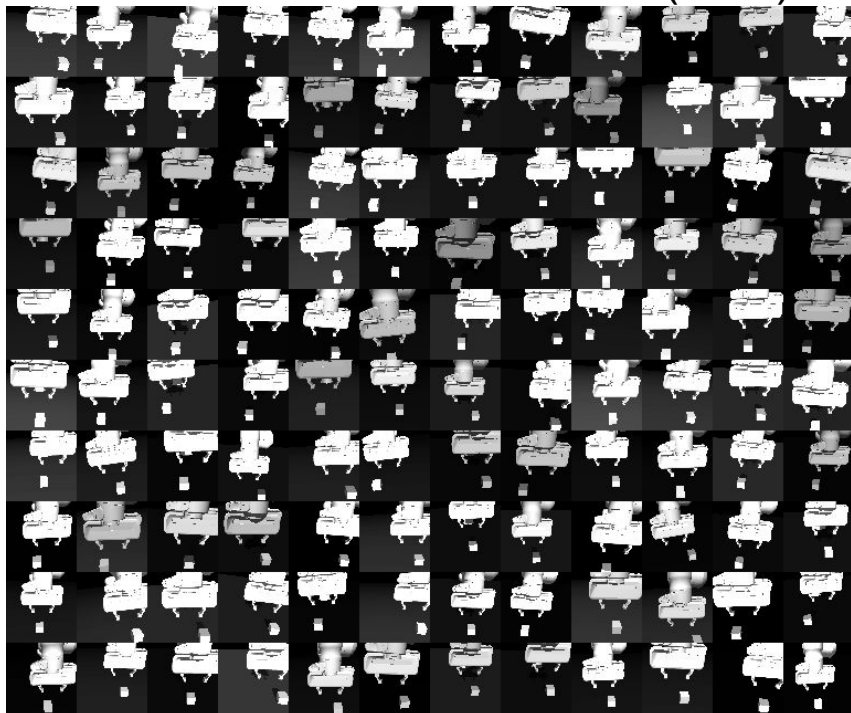
- Interested about physics of dynamical systems and how to control them.
- Enjoy working on most levels of the robotic stack
 - see a low-level controller executes a very complex behavior in synergy and precision.
 - Worked with a several real-world robots.

Real-world reinforcement learning



Sim-to-real

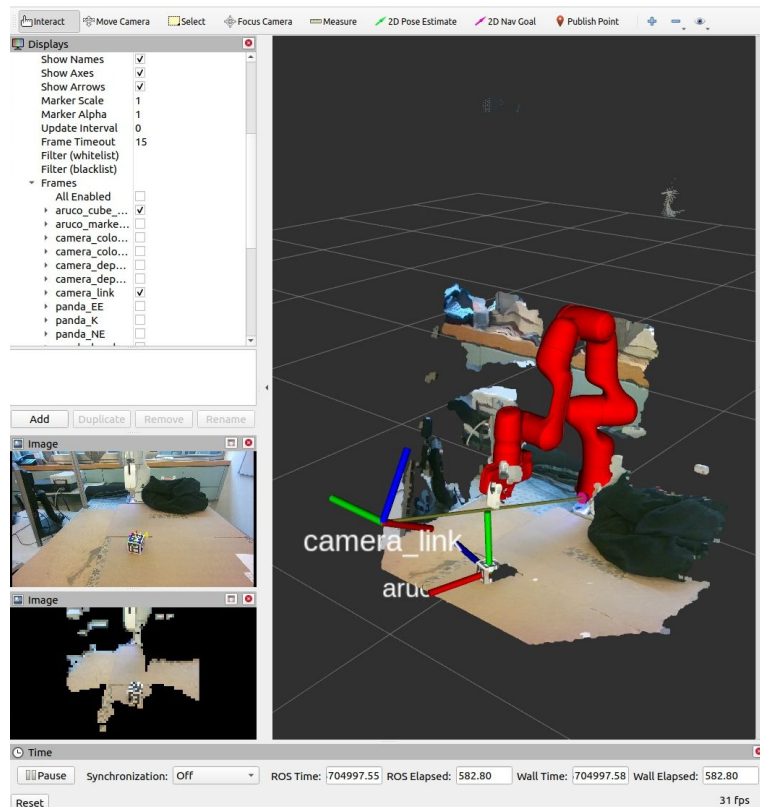
Sim: x512 environments (MJX)



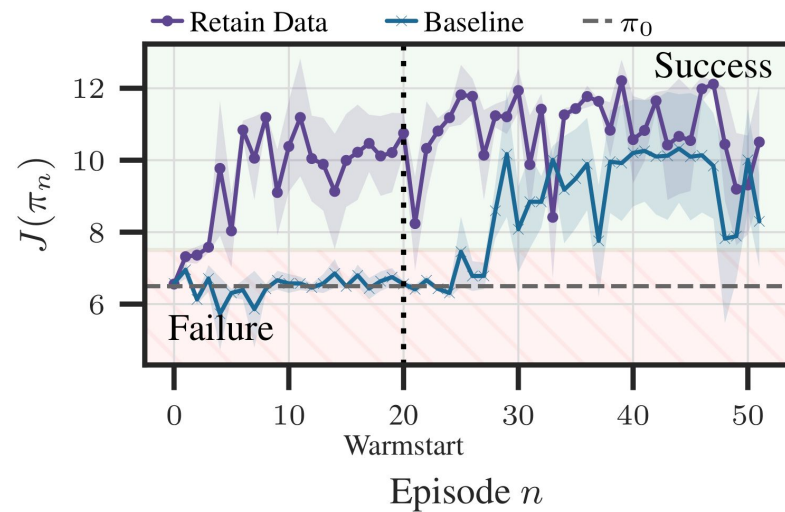
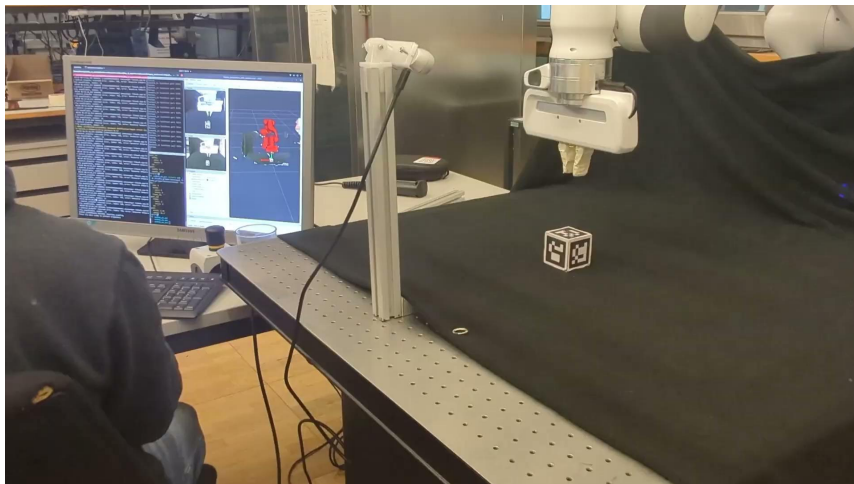
Real: x1 robot (!)



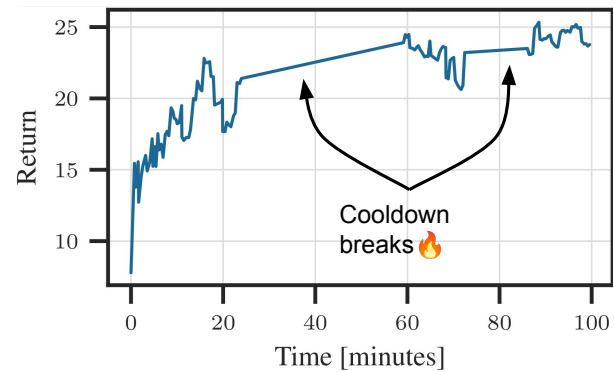
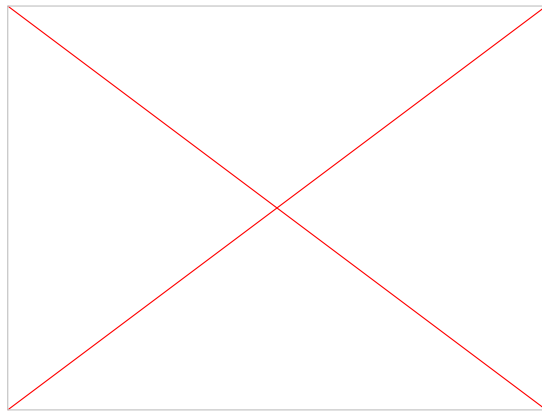
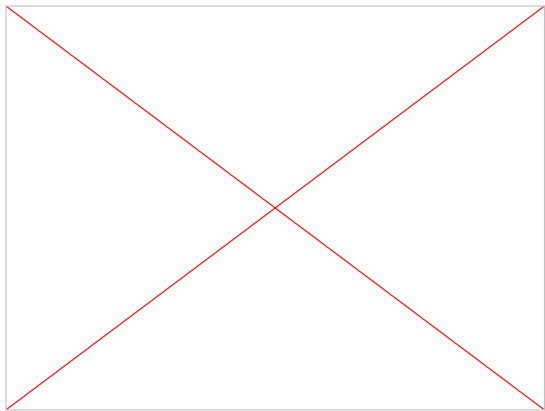
Visualization-first!



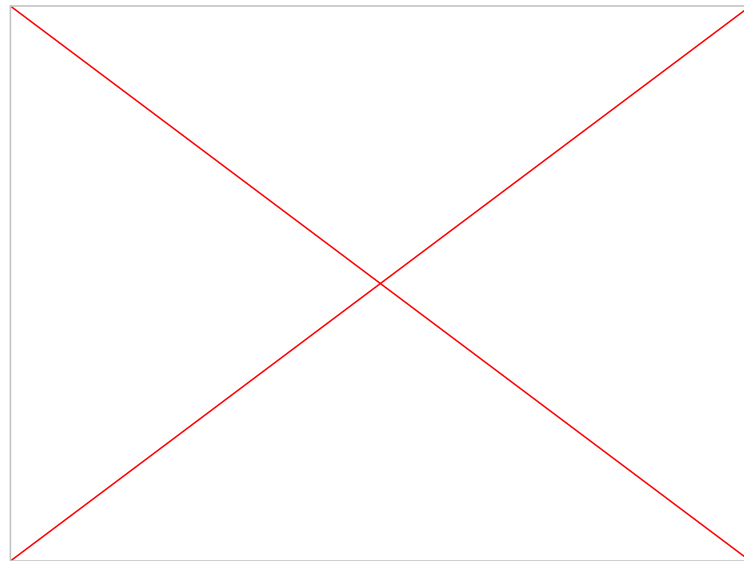
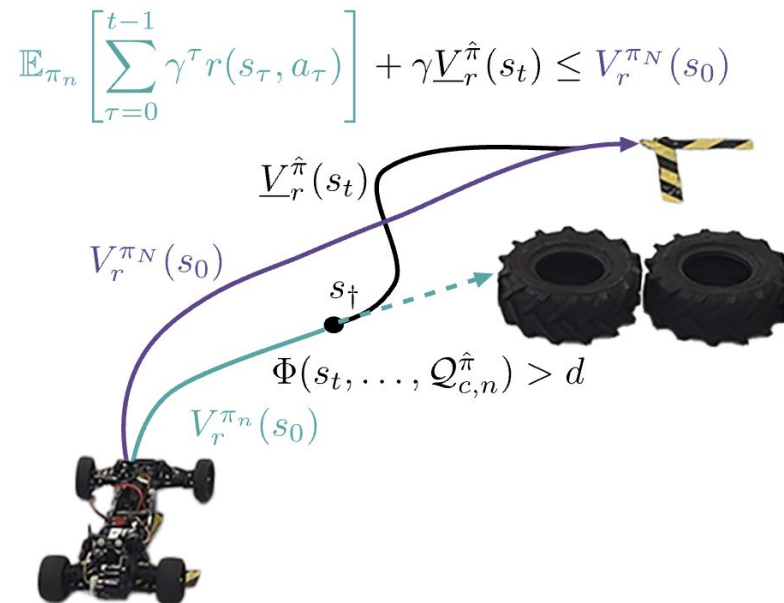
Online learning



Unitree Go2



SOOPER



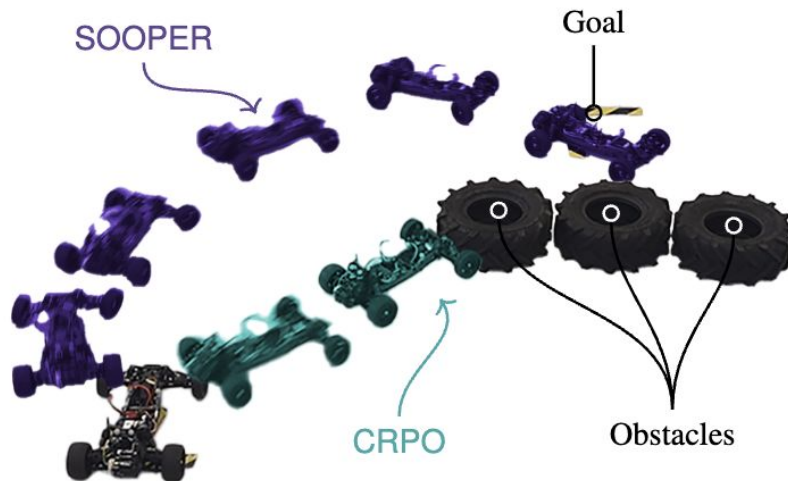
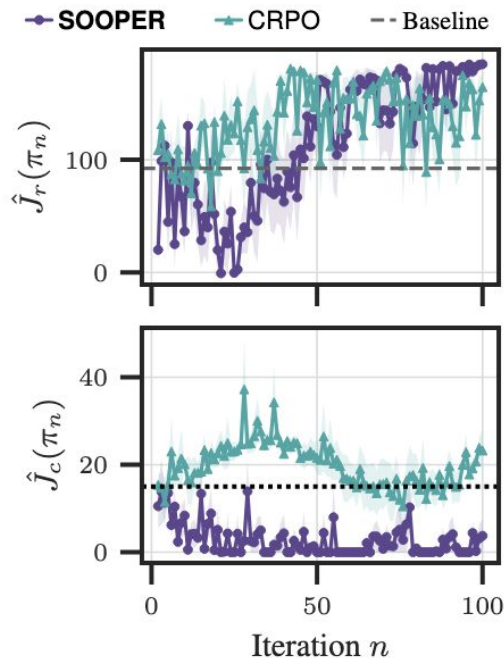
Theory

- Split episodic regret to a term that relates to performance in the simulated MDP and the performance of an optimal policy on the real MDP but under the augmented policy.
- First term: invoke theory for online learning in non-linear RKHS dynamics due to Kakade 2020.
- Second term: show that as model improves, the algorithm need not fall back to the prior policy

$$R(N) \leq \sum_{n=1}^N \underbrace{\left(J_r(\pi_c^*, f) - J_r(\bar{\pi}_{c,n}^*, f) \right)}_{\Delta_n^1 : \text{P.E.R. of } \bar{\pi}_c^* \text{ with } \Phi} + \sum_{n=1}^N \underbrace{\left(J_{\tilde{r}}(\pi^*, f) - J_{\tilde{r}}(\pi_n, f) \right)}_{\Delta_n^2 : \text{P.E.R. of } \pi_n \text{ on } \tilde{\mathcal{M}}}$$

Key result: SOOPER

- First of its kind result for cumulative regret for safe exploration
- Works under offline-to-online and model mismatch, scales to vision tasks.



Appendix & Misc

Libero -> MJX

